

The Execution Time of Basic Computational Operation τ (sec):

Accuracy = 1

Test Number	Matrix Size	Number of Iterations	Execution Time (sec)	Theoretical Time (sec)
1	10	26	0,000082	0,000011
2	100	41	0,008584	0,002573
3	1000	41	0,298175	0,266798
4	2000	41	1,083115	1,069330
5	3000	41	2,417052	2,407598
6	4000	41	4,281681	4,281602
7	5000	41	6,613727	6,691342
8	6000	41	9,635002	9,636818
9	7000	41	13,172069	13,118029
10	8000	41	17,034153	17,134976
11	9000	41	22,04987	21,687660
12	10000	41	27,363155	26,776079

The Results of the Computational Experiments for the Parallel Gauss-Seidel Algorithm

Parallel Program								
Grid size	Serial Algorithm	2 processors		4 processors		8 processors		
		Time	Speed Up	Time	Speed Up	Time	Speed Up	
10	0,000082	0,000801	0,102372	0,001153	0,071119	0,002793	0,029359	
100	0,008584	0,002993	2,868025	0,002543	3,375541	0,003047	2,817197	
1000	0,298175	0,14166	2,104864	0,077055	3,869639	0,20094	1,483901	
2000	1,083115	0,566288	1,912658	0,29728	3,643417	0,426056	2,542189	
3000	2,417052	1,266255	1,908819	0,677453	3,567852	0,979949	2,466508	
4000	4,281681	2,248605	1,90415	1,297455	3,300061	1,406907	3,043329	
5000	6,613727	3,48429	1,898156	1,95532	3,382427	2,063629	3,204901	
6000	9,635002	4,994915	1,928962	2,797522	3,44412	2,746721	3,50782	
7000	13,172069	6,817694	1,932042	3,920445	3,35984	3,925503	3,355511	
8000	17,034153	8,907715	1,912292	4,849285	3,512714	4,988497	3,414686	
9000	22,04987	11,40857	1,932746	6,613952	3,333842	6,128895	3,597691	
10000	27,363155	14,7205	1,858847	8,378713	3,265795	7,668384	3,568308	